

Background: Maternal and fetal health complications are widespread. One monitoring technique used during pregnancy is cardiotocography (CTG), which measures fetal heart rate (FHR) and uterine contractions (UC). This information can then be used to assess the well-being of the fetus with regards to hypoxia (oxygen deprivation). Machine learning is a tool that can assist clinicians to interpret CTGs, make the appropriate conclusions about fetal state (normal, suspect, pathologic), and ultimately provide timely interventions to reduce complications and infant and maternal mortality (Mehbodniya et al., 2022). Previous research has shown that tree-based (random forest, gradient boosting) methods perform the best on CTG data for fetal state predictions. Prior papers have also discussed use of oversampling suspect and pathologic minority classes to improve performance (Bhowmik et al., 2021; Mehbodniya et al., 2022; Nazli et al., 2025). We plan to build upon these findings by comparing different tree-based methods to identify critical variables in predicting fetal state and addressing procedures to handle fetal state class imbalance.

Description of dataset: The cardiotocography dataset was accessed through the UC Irvine Machine Learning Repository (Campos and Bernardes, 2000). The dataset has a sample size of 2126 mothers, with 21 features. Of these features, 19 will be used in the present analysis as predictors and 1 as an outcome variable (see last page for a table of feature descriptions).

Scientific objective: The scientific objective of this analysis is to classify fetal states using the CTG data. Fetal states can be classified into 3 levels: N for normal, S for suspect, and P for pathologic. Normal indicates a healthy heart rate pattern, S indicates some irregularities that may not necessarily indicate an issue but require further monitoring, and P indicates significant patterns of distress, such as lack of reactivity or prolonged decelerations. The majority class in this dataset is a normal fetal state. While previous methods have used tree-based methods for classification and have addressed class imbalances (Nazli et al., 2025), the present analysis will incorporate asymmetric punishment in the loss function for minority classes and adjust the probability thresholds for classification. The goal of this analysis is a final model with lower classification error and higher accuracy overall.

Methods we'll consider: To classify fetal states, we will compare tree-based methods (e.g., boosting, bagging) for predictive performance and variable importance. Trees are very interpretable models, which is important for clinical use by medical providers. Trees are also capable of capturing complex relationships and generally display high predictive performance, which could be instrumental in correctly classifying fetal states. Tree-based methods also allow for identification of the most important variables in predicting a fetal state, which can also help inform medical visits and further care.

Tuning parameters and evaluating model performance: To train the model, we will use an 80/20 train/test split. To account for the class imbalance, we will explore using oversampling of minority classes, asymmetric loss functions, and adjusting probability thresholds for classification. We will choose random forest hyperparameters (number of trees, tree depth, mtry) based on OOB error to guide choices. Boosting hyperparameters (number of trees, learning rate, tree depth) will be adjusted via cross validation.

Variables Used

NSP (<i>outcome variable</i>)	Fetal state code (normal, suspect, pathologic)
LB	Fetal heart rate (FHR) baseline (beats per minute)
FM	Number of fetal movements per second
UC	Number of uterine contractions per second
DL	Number of light decelerations per second
DS	Number of severe decelerations per second
DP	Number of prolonged decelerations per second
ASTV	Percentage of time with abnormal short term variability
MSTV	Mean value of short term variability
ALTV	Percentage of time with abnormal long term variability
MLTV	Median value of long term variability
Width	Width of fetal heart rate histogram
Min	Minimum of fetal heart rate histogram
Max	Maximum of fetal heart rate histogram
Nmax	Number of histogram peaks
Nzeroes	Number of zeros histogram zeros
Mode	Fetal heart rate histogram mode
Mean	Histogram mean
Median	Histogram median
Variance	Histogram variance
Tendency	Histogram tendency

References

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3. Nazli, I., Korbeko, E., Dogru, S., Kugu, E., & Sahingoz, O. K. (2025). Early detection of fetal health conditions using machine learning for classifying imbalanced cardiotocographic data. *Diagnostics*, 15(10), 1250. <https://www.mdpi.com/2075-4418/15/10/1250>
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